distribution as anticipated, illustrated in Figure 4 for *L-ViT* and in Appendix E.1 for BERT-Tiny. Equipped with this method, we now empirically validate the theoretical findings discussed in Section 4 by comparing transformers with smaller and larger minimum values of attention ORC, corresponding to increasing and decreasing variance. Our experimental results in Figure 5 indicate that increasing variance not only accelerates loss minimization but also leads to earlier convergence. The observations on text datasets, included in Appendix E.1, echo the same findings. This supports the implications of Lemma 4.2, which point to a link between ORC and the probability of exponential convergence of gradient descent, corresponding to slower or faster convergence. This finding also suggests that our proposed regularization can enhance the training of transformer models. In subsequent sections we further empirically demonstrate the impact of this curvature adjustment on transformers, effectively influencing the learning dynamics, performance, and robustness characteristics of the model.

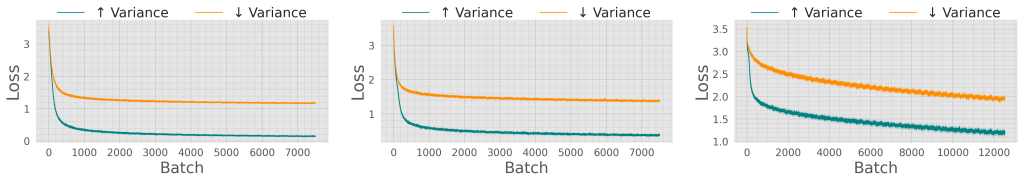

Figure 5: Training loss minimization on *L-ViT* with regularization methods increasing (↑) and decreasing (↓) variance for MNIST (left), Fashion-MNIST (middle), and CIFAR10 (right). Higher variance can promote faster loss reduction. The observations on text datasets, included in Appendix E.1, echo the same findings.

5 Robustness and Curvature Distribution in Transformers

Building on the insights from interpreting attention as a particle system as inroduced in Section 3 and using the well-established connection between ORC and system robustness (Bauer et al., 2017; Ollivier, 2009), we now leverage the geometry of the attention graph, specifically through the ORC perspective, to explain the robustness properties of transformers. We conduct a series of experiments exploring the relationship between ORC and robustness in transformers. The discussions in sections 2 and 3 suggest that more positive ORC values, should enhance the robustness of transformers. We empirically investigate this link through two focused studies: first, by examining the curvature and robustness of pre-trained transformers to input perturbation; and second, by assessing the impact of the ORC distribution on model forgetfulness. In line with the theoretical insight, our experiments confirm that more robust transformers feature a higher frequency of larger ORC values in their attention graphs. Additional experiments in Appendix E further support these findings.

Models and data. The experiments in this section and Appendix E are conducted on MNIST (Deng, 2012), Fashion-MNIST (Xiao et al., 2017), and CIFAR10 (Krizhevsky et al., 2009) images for the vision models, and on the IMDb Movie Reviews (Maas et al., 2011) (IMDb), Stanford Sentiment Treebank v2 (Socher et al., 2013) (SST-2) and the News Headlines Sarcasm Detection (Misra & Arora, 2023) (Headlines) datasets for the language models. We fine-tune pre-trained vision and language transformers on these datasets, comparing ViT (Dosovitskiy et al., 2020) against DeiT (Touvron et al., 2021) for vision, and BERT-Tiny (Turc et al., 2019) against ELECTRA-Small (Clark et al., 2020) for language models. All implementation details are reported in Appendix D.

Curvature and robustness of pre-trained transformers to input perturbation. We now compare the performance of fine-tuned models on perturbed and unperturbed test sets (further details in Appendix D). We quantify robustness by metric ρ,

$$\rho := \text{accuracy}_{\text{unperturbed}} - \text{accuracy}_{\text{perturbed}}. \tag{8}$$

We find that more positive-leaning ORC distributions correspond to smaller ρ values, indicating more robust models. For instance, ViT yields a ρ value of 0.067 on MNIST images, which is $\sim$20% smaller than the ρ value of 0.084 obtained from DeiT. Similarly, the ρ values for Fashion-MNIST and CIFAR10 datasets are $\sim$14% and $\sim$62% smaller in ViT than DeiT. Meanwhile, the ORC distribution in ViT is more positive-learning as quantified by the Wasserstein distance of the distribution from a δ

distribution at the ORC upper bound of 1, as demonstrated in Figure 6. This pattern holds in language transformers as well, where ELECTRA-Small exhibits a more positive-leaning ORC distribution than BERT-Tiny, and is $\sim 17\%$, $\sim 49\%$, and $\sim 23\%$ more robust to input perturbation on SST-2, IMDb, and Headlines datasets. Experiments on *L-ViT* and BERT-Tiny from random initialization (Appendix E.3) show a similar trend, where training a more robust model corresponds to eliminating negative ORC values leading to a distribution concentrated around a positive mode.

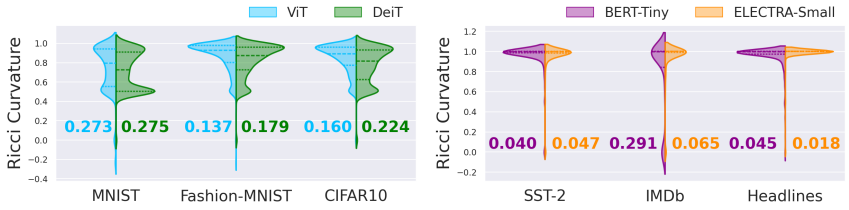

Figure 6: ORC distributions in the last attention layers of ViT and DeiT for a batch of MNIST, Fashion-MNIST, and CIFAR10 images and of BERT-Tiny and ELECTRA-Small for a batch of SST-2, IMDb, and Headlines texts. ViT (blue) and ELECTRA-Small (orange) exhibit more positive curvature values than DeiT (green) and BERT-Tiny (purple); quantified by the Wasserstein distances of the ORC distributions from a Dirac mass at the ORC upper bound of 1, annotated next to each distribution (smaller means more positive-leaning). Additional experiments in Appendix E further support these findings.

Attention curvature and forgetfulness in pre-trained transformers. Forgetfulness, where models lose previously learned information when exposed to new tasks or data, offers a complementary perspective on robustness, alongside robustness to input perturbation. Our experiments further link ORC in attention graphs to transformer robustness, showing that models with more positive-leaning ORC distributions are less prone to forgetfulness. We sequentially fine-tune each pre-trained model on a primary dataset and then a secondary one, tracking the performance decay on the primary dataset to measure how quickly each model "forgets" due to subsequent training. The results reveal a significant drop in classification accuracy –from above 80% to below 50%– highlighting a pronounced forgetting effect. DeiT forgets faster than ViT across all image dataset pairs (see Figure 7), and BERT-Tiny is more forgetful than ELECTRA-Small. Further details are in Appendix E.2. These findings align with our observations on perturbation robustness: ViT and ELECTRA-Small, with more positive-leaning ORC distributions and higher system robustness, show greater resistance to forgetting.

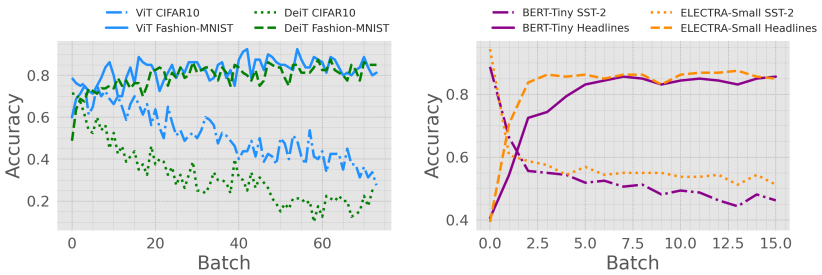

Figure 7: Test set accuracy of ViT (blue), DeiT (green) while fine-tuning on Fashion-MNIST after fine-tuning on CIFAR10 (left), and BERT-Tiny (purple) and ELECTRA-Small (orange) while fine-tuning on Headlines after fine-tuning on SST-2 (right). DeiT and BERT-Tiny demonstrate a faster degradation in performance compared to ViT and ELECTRA-Small due to subsequent fine-tuning. Further details and results are in Appendix E.

5.1 Robustness and Performance Trade-off

Our findings further highlight a potential contrast between the advantages of negative ORC for optimization and positive ORC for the robustness of transformers. The trade-off between robustness and performance is an established phenomenon studied in various contexts (Raghunathan et al., 2020; Xu & Mannor, 2006; Zhang et al., 2019). In light of the geometric underpinning of the

attention mechanism unveiled in this work, here we further explore this trade-off in the context of the distributional robustness of transformers, as indicated by their generalizability.

Table 1: Train and test accuracy and their gap using the increasing/decreasing variance regularization for six image and text datasets. Average and 95% confidence intervals over 30 trials are shown.

	Training accuracy (↑)		Test accuracy (↑)		Training accuracy - Test accuracy (↓)	
	Decrease variance	Increase variance	Decrease variance	Increase variance	Decrease variance	Increase variance
CIFAR10	**0.704 ± 0.001**	0.613 ± 0.009	**0.609 ± 0.002**	0.557 ± 0.005	0.095 ± 0.002	**0.056 ± 0.005**
Fashion-MNIST	**0.903 ± 0.000**	0.893 ± 0.001	**0.890 ± 0.001**	0.883 ± 0.001	0.013 ± 0.001	**0.010 ± 0.001**
MNIST	**0.981 ± 0.000**	0.980 ± 0.000	**0.984 ± 0.000**	0.982 ± 0.000	**-0.003 ± 0.001**	**-0.002 ± 0.000**
SST-2	**0.986 ± 0.001**	0.985 ± 0.001	0.826 ± 0.003	**0.831 ± 0.001**	0.160 ± 0.003	**0.154 ± 0.002**
IMDb	**0.994 ± 0.000**	0.993 ± 0.000	0.836 ± 0.003	**0.838 ± 0.002**	0.158 ± 0.003	**0.155 ± 0.002**
Headlines	0.965 ± 0.001	**0.967 ± 0.001**	**0.915 ± 0.002**	0.910 ± 0.002	**0.050 ± 0.002**	0.057 ± 0.001

We observe divergent impacts on performance and robustness from the opposing directions of variance regularization. Increasing the frequency of low-curvature edges by increasing variance speeds up optimization. Meanwhile, when this regularization introduces a second mode of highly positive curvature values, which is observed on all image datasets and two of the three text datasets in our experiments (Figure 4), it also improves generalizability, as evidenced by smaller train-test performance gaps. Conversely, decreasing variance enhances accuracy in most cases, while reducing generalizability. These effects are evident from Table 1. Our proposed regularization method globally impacts the attention scores and has limitations in targeted curvature adjustment, as, for instance, the simultaneous emergence of high and low curvature modes in some of our datasets suggests. Table 1 reports the results of the regularization methods, which together with the impact of each regularization on ORC distribution (figures 4 and 9), allow us to understand the impact of ORC distribution on accuracy and generalization. These results highlight the trade-off between better performance and robustness across different distributions, corresponding directly to the curvature distribution of the attention graph.

6 Related work

Related works are already discussed throughout the paper. In Appendix F, we provide a comprehensive literature review. Although the mentioned studies often serve as building blocks for our investigation, the link we establish in this paper between the geometry of attention maps, gradient descent-based training, performance, and robustness properties of transformers, has not been previously explored.

7 Conclusions, limitations, and ethical considerations

We presented an analysis of transformers through the lens of graph geometry, showing that the Ricci curvature of attention maps is a key factor in training dynamics and robustness of transformers. Our results establish a connection between the Ricci curvature of the attention graph and the convergence behavior of gradient descent in transformers. Meanwhile, we show that more positive attention curvature values correspond to more robust transformers. Our findings also indicate a trade-off between performance and robustness. Informed by these findings, we introduced an efficient curvature-adjusted regularization method that allows us to improve training or generalizability in transformers. Furthermore, we examine transformers' reliance on attention edges associated with different curvature values.

Limitations. A potential limitation of this work lies in the inherent difficulty of theoretical analysis of transformer models employed in practice and direct empirical validation of theoretical results derived in simplified settings. While our theory employs idealized transformers with simplifying assumptions, such simplifications are well-established traditions, due to the near intractability of fully characterizing practical transformer models. Furthermore, our empirical results are well-aligned with our theoretical discussion, confirming that our theory can extend to practical settings.

Ethical considerations. Our paper is theoretical and foundational in nature, but addresses the success of transformers, a family of machine learning models that has significant ethical considerations which are well-documented and are part of active research in the community.